

BANK LOAN ANALYTICS REPORT

SQL + Power BI Portfolio Project

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Applications 38,576	Funded amount 435.8M	Amount received 473.1M
Average interest 12.05%	Average DTI 13.33%	Charged-off rate 13.82%

This report is based on the supplied **financial_loan.csv**, the supplied SQL scripts and the supplied Power BI dashboard screenshots. The figures below are calculated from the dataset so that the report stays consistent with the actual project.

The report covers portfolio volume, funding and repayment, monthly trend, credit grade, purpose, term, verification, state exposure and the business recommendations supported by those results.

Note: The report intentionally does not include a Future Improvements section.

1. Project Overview

The project analyzes the bank loan portfolio using SQL for data exploration and business questions, then uses Power BI to present the results in an interactive two-page dashboard.

Project workflow

Stage	Work completed	Output
Data understanding	Checked structure, record count, duplicates, missing values and dates.	Data-quality baseline
SQL EDA	Reviewed status, purpose, grade, sub-grade, ownership, verification, term and state.	Portfolio profile
Business questions	Calculated volume, funding, repayments, risk rates and monthly trends.	Business findings
Power BI	Used the supplied Summary and Overview pages for interactive reporting.	Dashboard

Dataset

Item	Value
Records	38,576
Columns	24
Date range	01 Jan 2021 to 12 Dec 2021
Loan statuses	Fully Paid, Charged Off, Current
Core numeric fields	ANNUAL_INCOME, DTI, INSTALLMENT, INT_RATE, LOAN_AMOUNT, TOTAL_PAYMENT

The supplied SQL table definition contains the loan, borrower, status, income, DTI, interest-rate, loan-amount and payment fields used in this report.

2. Data Quality & SQL Work

I started the analysis with basic checks before interpreting the portfolio. The purpose was to confirm the size of the dataset and identify obvious duplication or missing-data issues.

Check	Result	Meaning
Total records	38,576	Portfolio size used throughout the report
Duplicate Loan IDs	0	No repeated loan IDs found
Duplicate Member IDs	0	No repeated member IDs found
Duplicate complete rows	0	No identical duplicate rows found
Missing cells	1,438	Missing values are present in EMP_TITLE
Missing EMP_TITLE	1,438	Employment-title field is incomplete for some records

SQL analysis areas

Area	Examples
Portfolio size	Total applications, total loan amount, total amount received
Loan status	Fully Paid, Charged Off and Current
Risk	Charged-off loans, bad-loan rate by grade and purpose
Loan structure	Term, home ownership and verification
Geography	State-wise applications and loan amount
Time	Monthly applications, loan amount and payment
Credit profile	Average interest rate and DTI by grade

The SQL scripts explicitly include duplicate checks, missing-value checks, date range checks, status distribution, grade/purpose/term analysis and business questions around bad-loan rates.

3. Portfolio Snapshot

These figures are calculated from the supplied dataset and align with the KPI values shown in the Power BI screenshots.

Loan applications	Funded amount	Amount received
38,576	435.8M	473.1M
Average interest	Average DTI	Payment difference
12.05%	13.33%	37.3M

Loan status breakdown

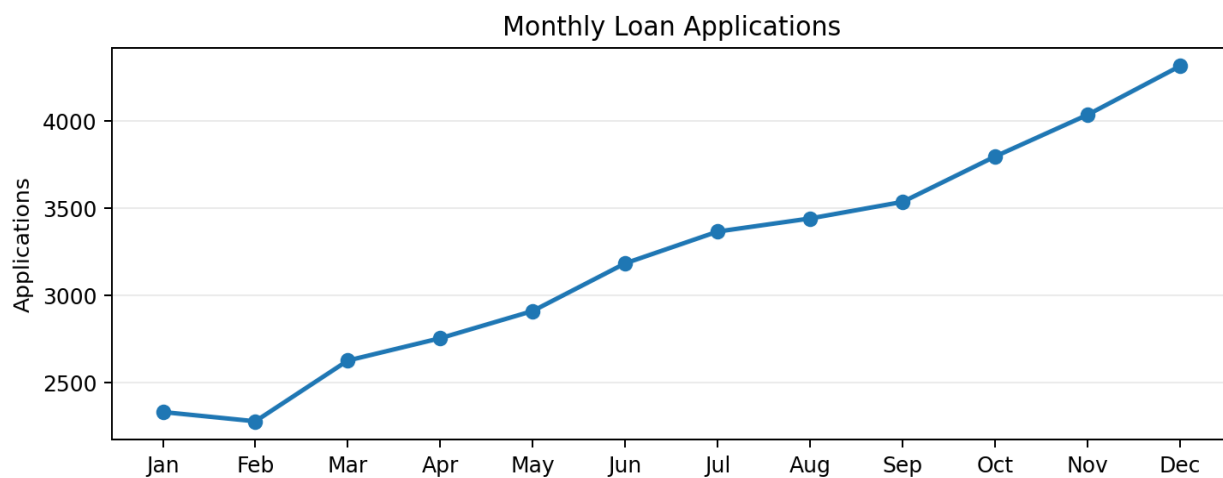
Status	Applications	Share	Loan amount	Amount received
Fully Paid	32,145	83.33%	351.4M	411.6M
Charged Off	5,333	13.82%	65.5M	37.3M
Current	1,098	2.85%	18.9M	24.2M

Dashboard Good/Bad definition: Good = Fully Paid + Current = 33,243 applications (86.18%) and 370.2M funded. Bad = Charged Off = 5,333 applications (13.82%) and 65.5M funded.

The SQL business-question script also keeps Current as a separate category. That is why a SQL Good/Bad/Current table and the dashboard Good/Bad donut should not be interpreted as the same grouping.

4. Monthly Portfolio Trend

Applications rise from 2,332 in January to 4,314 in December, an increase of 85.0%. Monthly loan amount rises from 25.0M to 54.0M, an increase of 115.7%.

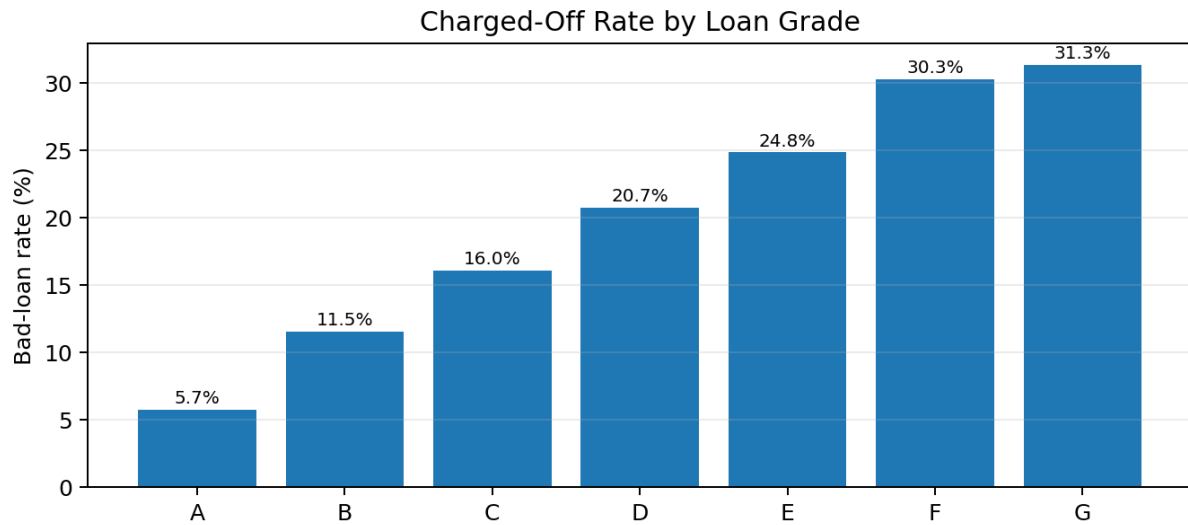


Month	Applications	Loan amount	Amount received
Jan	2,332	25.0M	27.6M
Feb	2,279	24.6M	27.7M
Mar	2,627	28.9M	32.3M
Apr	2,755	29.8M	32.5M
May	2,911	31.7M	33.8M
Jun	3,184	34.2M	36.2M
Jul	3,366	35.8M	38.8M
Aug	3,441	38.1M	42.7M
Sep	3,536	40.9M	44.0M
Oct	3,796	44.9M	49.4M
Nov	4,035	47.8M	50.1M
Dec	4,314	54.0M	58.1M

December has the highest application count and the highest loan amount in the dataset. The increase is gradual across the year rather than being driven by one isolated month.

5. Credit Risk by Grade

The strongest risk pattern in the dataset is the increase in charged-off rate across credit grades. Average interest rate also increases as the grade moves from A to G.



Grade	Applications	Bad loans	Bad rate	Avg. interest
A	9,689	552	5.70%	7.35%
B	11,674	1,343	11.50%	11.03%
C	7,904	1,266	16.02%	13.55%
D	5,182	1,072	20.69%	15.71%
E	2,786	691	24.80%	17.71%
F	1,028	311	30.25%	19.74%
G	313	98	31.31%	21.40%

Grade A has a 5.70% bad-loan rate; Grade G has 31.31%. Average interest rises from 7.35% to 21.40%.

These are observed portfolio patterns. The analysis does not establish that grade or interest rate by itself causes a loan to be charged off.

6. Loan Purpose Analysis

Purpose volume and purpose-level risk are not the same thing. Debt consolidation dominates the portfolio by application count, while small business has the highest observed bad-loan rate.

Purpose	Applications	Bad loans	Bad rate	Charged-off amount
Debt Consolidation	18,214	2,651	14.55%	36.3M
Credit Card	4,998	508	10.16%	6.7M
Other	3,824	587	15.35%	5.0M
Home Improvement	2,876	327	11.37%	3.9M
Major Purchase	2,110	206	9.76%	1.9M
Small Business	1,776	455	25.62%	6.7M
Car	1,497	155	10.35%	1.1M
Wedding	928	86	9.27%	1.0M
Medical	667	100	14.99%	0.9M
Moving	559	84	15.03%	0.5M
House	366	57	15.57%	0.7M
Vacation	352	50	14.20%	0.3M
Educational	315	50	15.87%	0.4M
Renewable Energy	94	17	18.09%	0.2M

Debt consolidation has 18,214 applications and 36.3M of charged-off loan amount. Small business has 1,776 applications and the highest observed bad-loan rate at 25.62%.

7. Loan Term & Verification

The two loan-term segments have noticeably different observed bad-loan rates.

Loan term

Term	Applications	Bad loans	Bad rate
36 months	28,237	3,023	10.71%
60 months	10,339	2,310	22.34%

The 60-month segment has a 22.34% bad-loan rate versus 10.71% for 36 months.

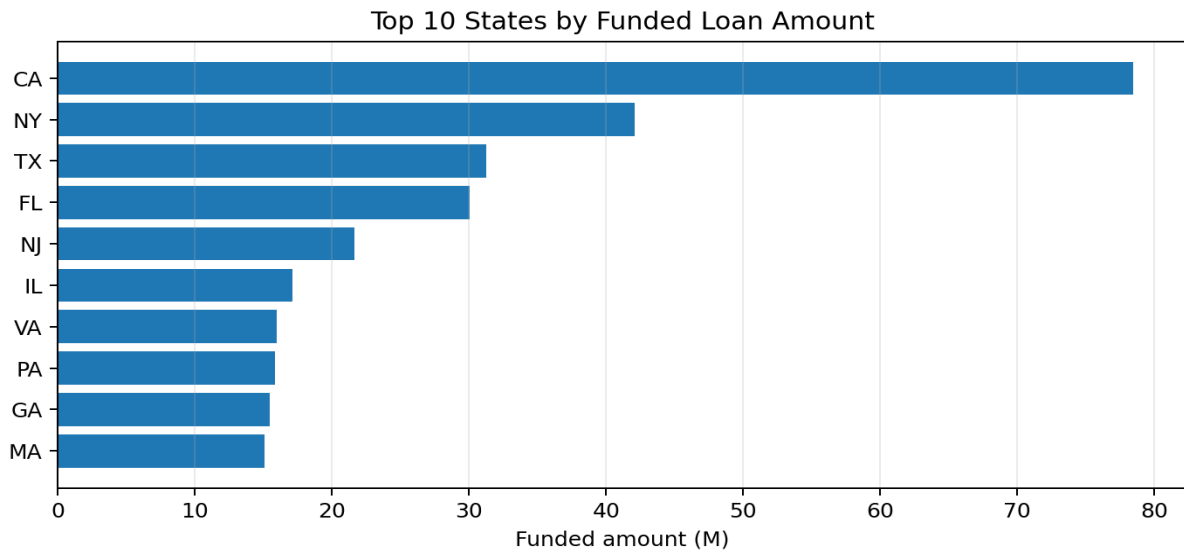
Verification status

Verification status	Applications	Bad loans	Bad rate
Not Verified	16,464	2,015	12.24%
Source Verified	9,777	1,382	14.14%
Verified	12,335	1,936	15.70%

The highest observed bad-loan rate among the verification groups is 15.70% for Verified. This is a descriptive comparison, not a causal conclusion.

8. State-Level Portfolio Exposure

State analysis shows where the largest application volumes and funded amounts are concentrated.

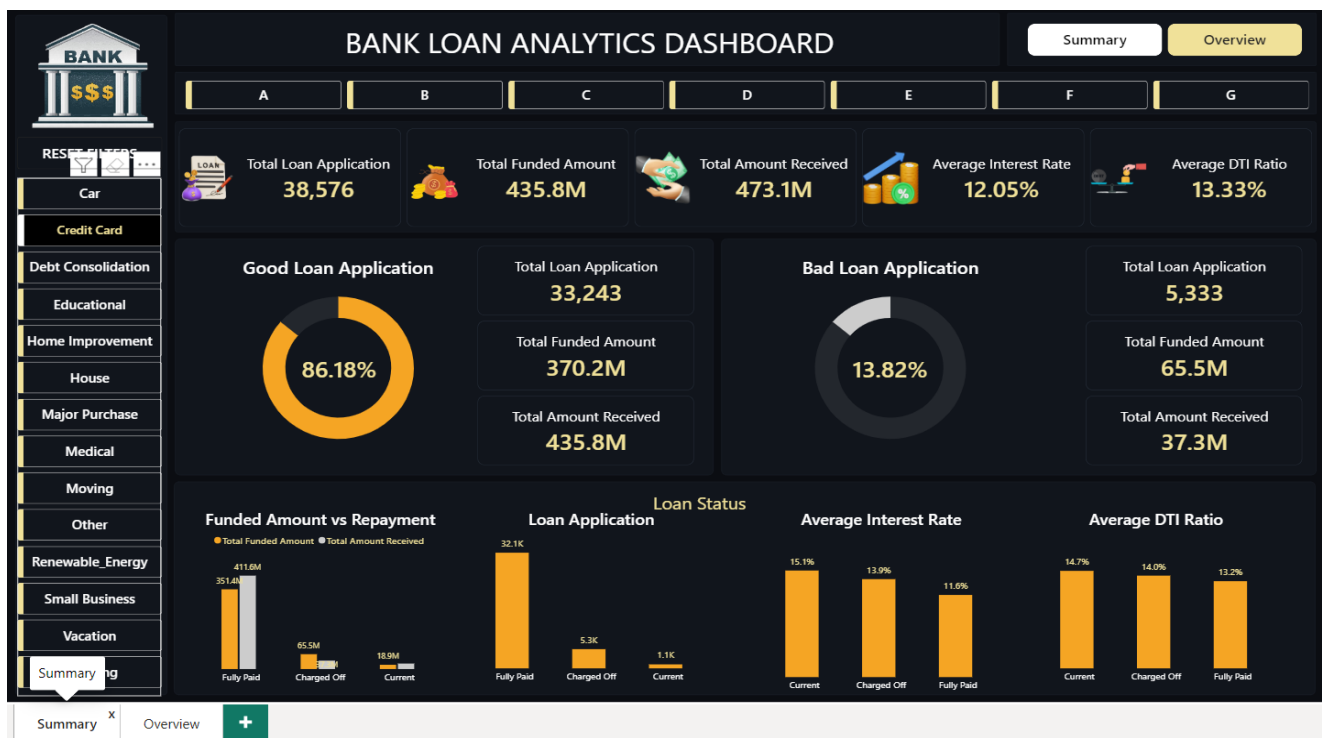


State	Applications	Funded amount	Bad loans	Bad rate
CA	6,894	78.5M	1,057	15.33%
NY	3,701	42.1M	468	12.65%
TX	2,664	31.2M	301	11.30%
FL	2,773	30.0M	479	17.27%
NJ	1,822	21.7M	273	14.98%
IL	1,486	17.1M	192	12.92%
VA	1,375	16.0M	170	12.36%
PA	1,482	15.8M	169	11.40%
GA	1,355	15.5M	207	15.28%
MA	1,310	15.1M	150	11.45%

California is the largest state by funded exposure with 78.5M. It also has 13.1M in charged-off loan amount. Florida's observed bad-loan rate is 17.27%, compared with California's 15.33%.

9. Power BI Summary Page

The supplied Summary page is focused on portfolio health and loan-status analysis.

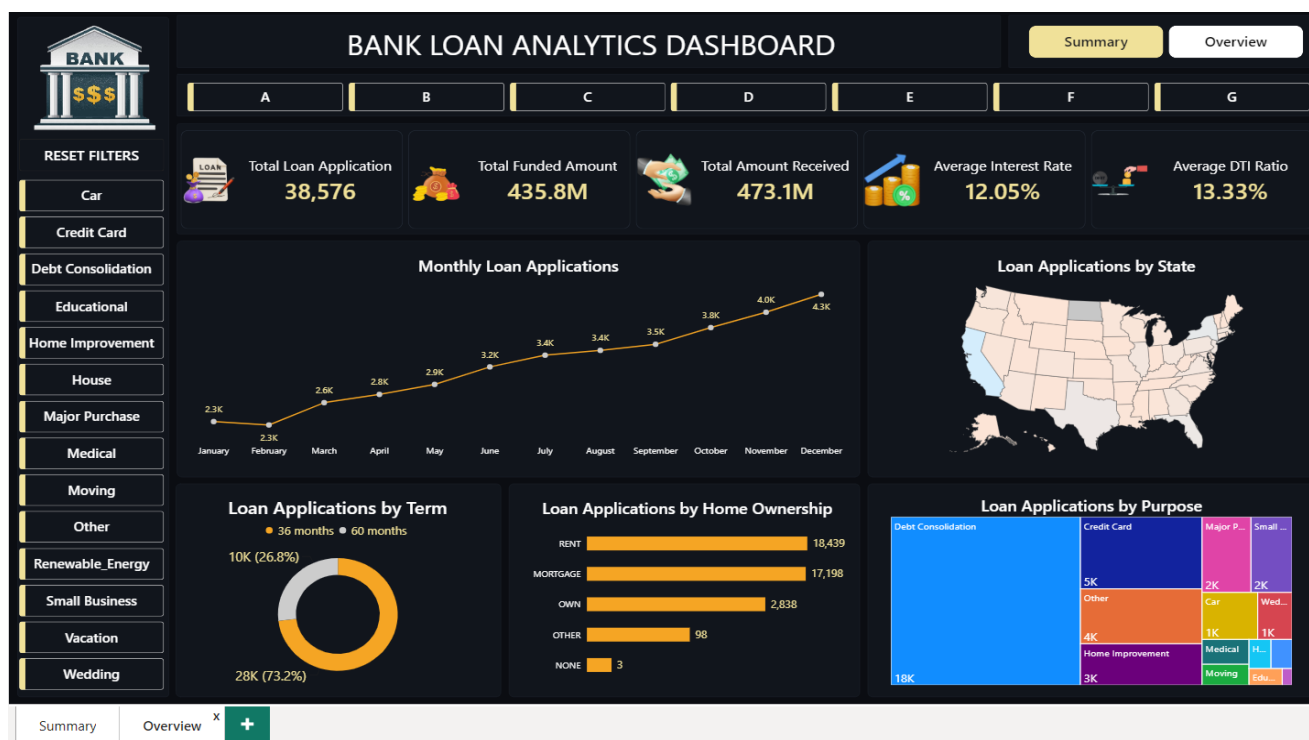


KPI cards: Total Loan Application, Total Funded Amount, Total Amount Received, Average Interest Rate and Average DTI Ratio.

Main visuals: Good Loan Application, Bad Loan Application, Funded Amount vs Repayment by Loan Status, Loan Application by status, Average Interest Rate by status and Average DTI Ratio by status.

10. Power BI Overview Page

The supplied Overview page focuses on volume, geography and portfolio segmentation.



Main visuals: Monthly Loan Applications, Loan Applications by State, Loan Applications by Term, Home Ownership and Loan Purpose.

The dashboard also includes purpose/grade filters, page navigation and a reset-filters control.

SQL-to-dashboard connection

Dashboard view	Related SQL analysis
Monthly applications	DATE_TRUNC month + COUNT(*)
State analysis	State-wise COUNT(*) and SUM(LOAN_AMOUNT)
Term	36 months vs 60 months
Purpose	Purpose-wise application distribution
Good/Bad	Loan status grouping and charged-off analysis
Risk	Bad-loan rate by grade, purpose, term and verification

11. Business Findings & Recommendations

These recommendations are based on the observed patterns in the dataset. They are framed as monitoring or review actions rather than claims about cause and effect.

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|----|--|
| 01 | Monitor higher-risk grades
Grades E, F and G have bad-loan rates of 24.80%, 30.25% and 31.31%. Track these grades separately in regular portfolio reviews. |
| 02 | Review 60-month lending separately
The 60-month segment shows a 22.34% bad-loan rate versus 10.71% for 36 months. Compare the two segments regularly. |
| 03 | Watch small-business exposure
Small business has 1,776 applications and a 25.62% bad-loan rate. Monitor both the rate and the absolute number of charged-off loans. |
| 04 | Track large state exposures
California has 78.5M of funded exposure and 13.1M charged-off amount. High-exposure states deserve regular portfolio monitoring. |
| 05 | Separate volume from risk
Debt consolidation is the largest purpose with 18,214 applications, but small business has the highest observed bad-loan rate. Use both volume and risk measures. |
| 06 | Review funding and repayment together
The portfolio shows 435.8M funded and 473.1M received. The 37.3M aggregate difference should not automatically be described as profit; it is simply the difference between the two totals. |

12. Final Takeaways

The project brings together SQL analysis and Power BI reporting into one practical loan-portfolio analysis.

- 1** The dataset contains 38,576 applications, 435.8M funded and 473.1M received.
- 2** Charged Off represents 5,333 applications (13.82%). In the supplied dashboard, Fully Paid + Current are grouped as Good, giving 33,243 applications (86.18%).
- 3** Applications rise from 2,332 in January to 4,314 in December, while monthly loan amount rises from 25.0M to 54.0M.
- 4** Observed bad-loan rate rises from 5.70% in Grade A to 31.31% in Grade G.
- 5** The 60-month segment has a 22.34% bad-loan rate versus 10.71% for 36 months.
- 6** Debt consolidation is the largest purpose by applications; small business has the highest observed purpose-level bad-loan rate.
- 7** California has the largest funded exposure among states in this dataset.

Project positioning: SQL was used for data validation, exploration and business questions; Power BI was used to communicate the results through an interactive portfolio dashboard.

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